Assignment 3 (v2)

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Q1. Create a trigger on Table of your choice to check if the Primary key ID already exists or not , before inserting a new record. & Send a custom reply instead of an error message.

Ans)

Function:

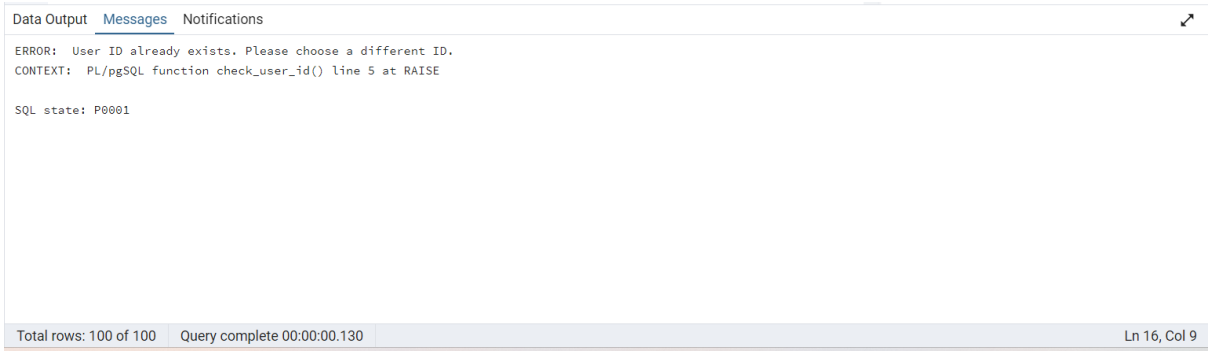
```
CREATE OR REPLACE FUNCTION check_user_id()

RETURNS TRIGGER

LANGUAGE plpgsql

AS $BODY$

BEGIN

   IF EXISTS (SELECT 1 FROM "EC_DB".users WHERE user_id = NEW.user_id)
THEN

      RAISE EXCEPTION User ID already exists. Please choose a different ID.';

   END IF;

   RETURN NEW;

END;

$BODY$;
```

Trigger:

```
CREATE OR REPLACE TRIGGER trigger_q1

BEFORE INSERT ON "EC_DB".users

FOR EACH ROW

EXECUTE FUNCTION check_user_id();
```

TestCase:

insert into "EC_DB".users values
(1,'tmp','tmp@gamil.com','asdfg','tmp','tmp','tmp');



Q2. Create a trigger on the Table of your choice to check if the Foreign key ID already exists or not before inserting a new record. & Send a custom reply instead of an error message.

Ans)

Function:

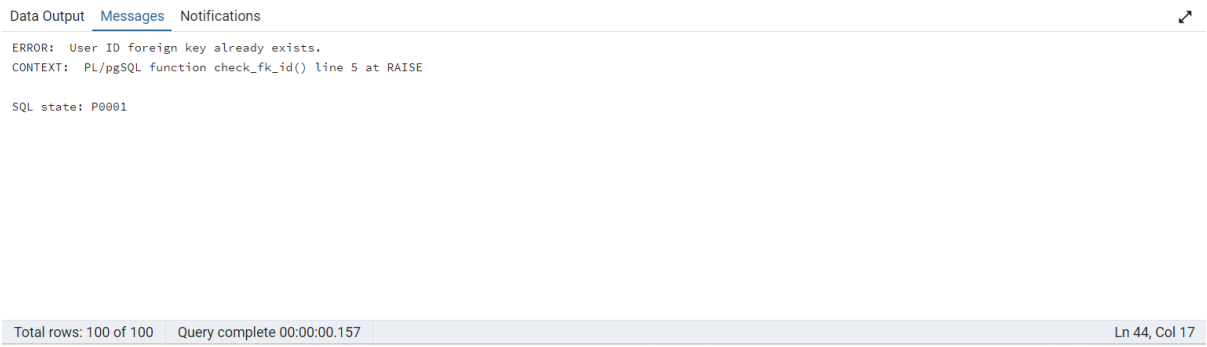
```
CREATE OR REPLACE FUNCTION check_fk_id()
RETURNS TRIGGER
LANGUAGE plpgsql
AS $BODY$
BEGIN
    IF EXISTS (SELECT 1 FROM "EC_DB".orders WHERE user_id = NEW.user_id)
THEN
       RAISE EXCEPTION 'User ID foreign key already exists.';
   END IF;
   RETURN NEW;
END;
$BODY$;
```

Trigger:

```
CREATE OR REPLACE TRIGGER trigger_q2
BEFORE INSERT ON "EC_DB".orders
FOR EACH ROW
EXECUTE FUNCTION check_fk_id();
```

TestCase:

```
insert into "EC_DB".orders values (101,39,'24-07-25','abcdefgh',2525.5,'Pending');
```



Q3. Write a SQL function that takes a product's product id and a discount percentage as inputs and returns the discounted price Of the product.

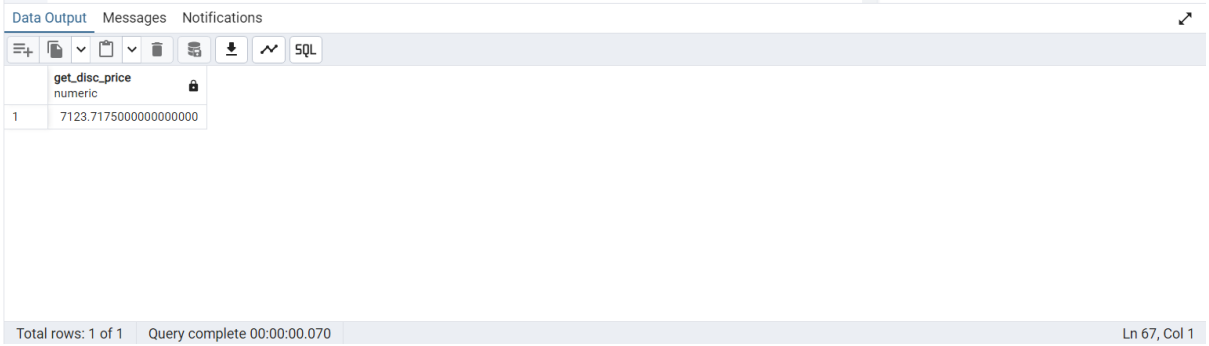
Ans)

Function:

```
CREATE OR REPLACE FUNCTION get_disc_price(pro_id INT, disc_per NUMERIC)
RETURNS NUMERIC
LANGUAGE plpgsql
AS $BODY$
DECLARE
    og_price NUMERIC;
    disc_price NUMERIC;
BEGIN
```

```
    SELECT price INTO og_price
    FROM "EC_DB".products
    WHERE product_id = pro_id;

    disc_price = og_price - (og_price * disc_per / 100);

    RETURN disc_price;
END;
$BODY$;
```

TestCase: select get_disc_price(1,5);



Q4. Create a stored procedure named add_order that takes the user_id, shipping_address, and a list of product_id and quantity pairs as inputs, and inserts a new order into the orders and order_details tables. The procedure should also update the stock quantity of the products.

Ans) Didn't understand how to do it

Q5. Write a trigger that automatically decreases the stock quantity of a product in the products table when a new order is inserted into the order details table.

Ans)

Function:

```
CREATE OR REPLACE FUNCTION dec_stock_qty()
RETURNS TRIGGER
LANGUAGE plpgsql
AS $BODY$
BEGIN

   UPDATE "EC_DB".products
   SET stock_quantity = stock_quantity - NEW.quantity
   WHERE product_id = NEW.product_id;
   RETURN NEW;
END;
$BODY$;
```

Trigger:

```
CREATE OR REPLACE TRIGGER trigger_q5
AFTER INSERT ON "EC_DB".order_details
FOR EACH ROW
EXECUTE FUNCTION dec_stock_qty();
```

Test Case:

```
INSERT INTO "EC_DB".ORDER_DETAILS VALUES (302,100,50,10,6000)
select * from "EC_DB".order_details;
```

Data Output | Messages | Notifications

	order_detail_id [PK] integer	order_id integer	product_id integer	quantity integer	price numeric (10,2)
294	294	83	48	7	619.66
295	295	52	18	5	1460.92
296	296	97	45	2	3303.66
297	297	43	24	4	7498.65
298	298	56	30	9	6725.63
299	299	14	32	4	4643.25
300	300	2	21	7	7900.81
301	301	100	1	10	5000.00
302	302	100	50	10	6000.00

Total rows: 302 of 302 | Query complete 00:00:00.084 | Ln 91, Col 1

Data Output | Messages | Notifications

	product_id [PK] integer	name character varying (100)	description text	price numeric (10,2)	stock_quantity integer	category_id integer
42	43	Car Battery	Power source for starting and operating a vehicle.	8287.20	66	9
43	44	Bike Helmet	Protective headgear for cycling safety.	3346.47	52	9
44	45	Car Vacuum Cleaner	Portable vacuum for cleaning vehicle interiors.	6725.63	24	9
45	46	Sofa	Comfortable seating for multiple people.	9818.68	62	10
46	47	Dining Table	Table used for meals in a dining area.	3303.66	32	10
47	48	Chair	Single-seat furniture for sitting.	167.62	50	10
48	49	Bed	Furniture for sleeping and resting	2431.66	97	10
49	1	Mobile Phone	Portable communication device with internet access and app...	7498.65	18	1
50	50	Wardrobe	Storage unit for clothes and accessories.	2743.24	33	10

(Qantity Reduced)